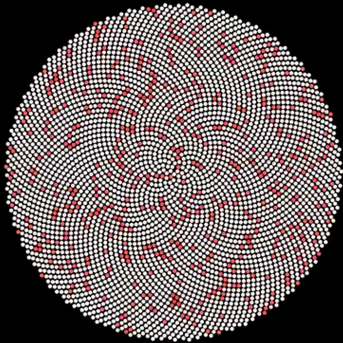


Cedric Allier

# Plexus

# Gray-Scott chemistry on a disc of 4,000 cells



## entities

cell 4,000

## activities



seed

the initial positions

the initial morphogen



rewire

neighbours within a radius



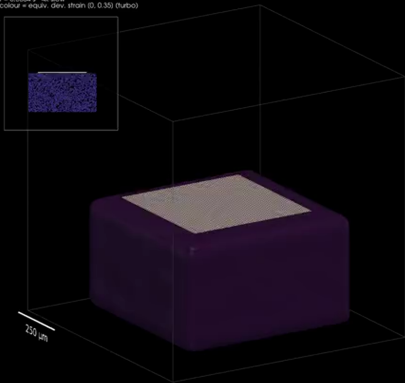
lateral

morphogen diffusion between neighbouring cells

autocatalytic reaction

# An elastic sheet pressed into a gel of 500,000 material points

mesh mpm\_sheet\_elastic box 1 mm cube  
500,000 mpm\_particles grid 64x64 dx 15.6  $\mu$ m  
frame 2/400 275 ms/frame compute  
t = 0.0064 s 4x slow  
colour = equiv. dev. strain (0, 0.35) (turbo)



## entities

|              |         |                 |
|--------------|---------|-----------------|
| vertex       | 4,096   |                 |
| ↔ half_edge  | 18,432  | vertex → vertex |
| plate_face   | 2,304   |                 |
| mpm_particle | 500,000 | gel             |
| mpm_grid     |         | field, grid 64  |

## activities

-  seed  
lay out the plate
-  structural  
drive the surface at a set rate
-  lateral  
contact with a live surface  
the material update
-  exchange  
particle to grid  
grid to particle
-  field  
the grid solve

# 60,000 particles riding the Halvorsen attractor

## entities

cloud 60,000 c

## activities

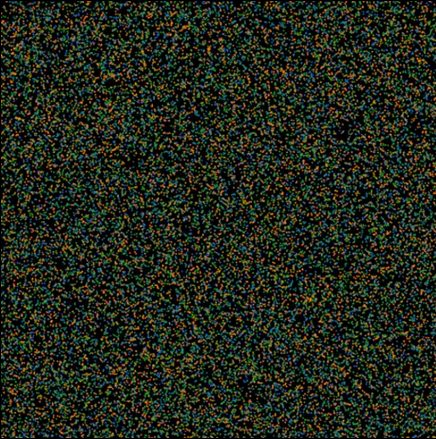


lateral

flow along a strange attractor



# Three particle types under one pairwise force law



## entities

particle 30,000  $t_0$ ,  $t_1$ ,  $t_2$

## activities



rewire

neighbours within a radius



lateral

attraction and repulsion

# A vesicle that grows petals where the activator is high



## entities

|             |         |                    |
|-------------|---------|--------------------|
| cell        | 2,000   | of 65,004 declared |
| vertex      | 130,004 |                    |
| ↔ half_edge | 520,032 | vertex → vertex    |

## activities



seed

build the cell mesh  
the initial morphogen



aggregate  
cell geometry from the mesh



rewire  
who neighbours whom  
T1 neighbour exchange



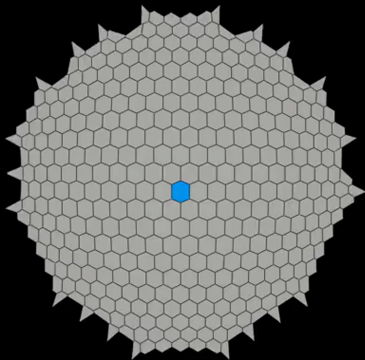
lateral  
morphogen diffusion between neighbouring cells  
autocatalytic reaction  
chemistry driven by curvature  
growth where the morphogen is high  
vertex-model shape energy  
purse-string line tension



divide  
cell division

# An epithelial sheet of 400 cells extruding its dying cells

sheet\_die\_mesh frame 1/60 374 cells



## entities

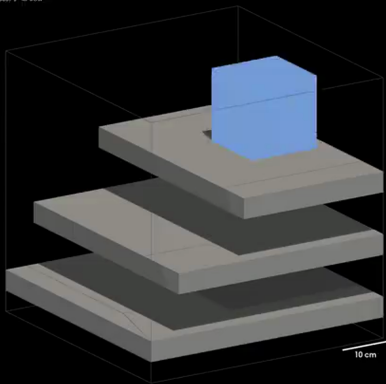
cell 400 of 2,000 declared  
vertex 4,000  
↔ half\_edge 16,000 vertex → vertex

## activities

- seed  
build the cell mesh
- aggregate  
cell geometry from the mesh
- lateral  
vertex-model shape energy
- die  
cell death
- rewire  
T1 neighbour exchange

# A waterfall of 5,000,000 material points

if waterfall\_5m box 50 cm cube  
5,000,000 particles grid 192x3 = 2.1M cells, dx 2.6 mm  
frame 8/2530 3386 m/frame compute  
t = 0.006667 s dr slow



## entities

|                |           |                    |
|----------------|-----------|--------------------|
| cell           | 1         | water              |
| ↔ mpm_particle | 5,000,000 | 5,000,000 per cell |
| mpm_grid       |           | field, grid 192    |

## activities



lateral  
gravity  
the material update  
shear viscosity

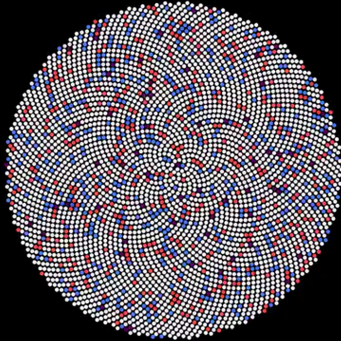


exchange  
particle to grid  
grid to particle



field  
the grid solve

# Two coupled chemistries on a disc of 4,000 cells



## entities

cell 4,000

## activities



seed

the initial positions

the initial morphogen

the initial morphogen



rewire

neighbours within a radius



lateral

morphogen diffusion between neighbouring cells

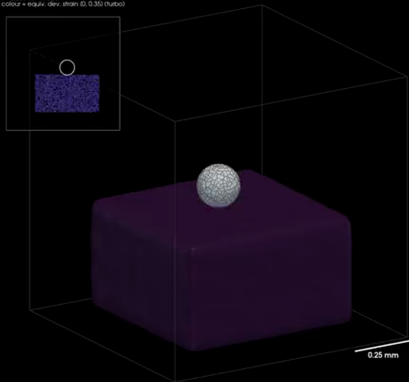
morphogen diffusion between neighbouring cells

morphogen diffusion between neighbouring cells

autocatalytic reaction

# A spheroid of 400 cells inflating inside a gel

mesh\_mpm\_spheroid\_gel\_box 1 mm cube  
500,000 particles grid 64x64 = 2048 cells, dx 0.0156 mm  
frame 2,400 - 53 ms/frame compute  
t = 0.0264 s Ax slow  
colour = eqv. dev. strain (0, 0.35) (turbo)



## entities

|              |         |                 |
|--------------|---------|-----------------|
| vertex       | 4,096   |                 |
| ↔ half_edge  | 4,096   | vertex → vertex |
| sphere_face  | 512     |                 |
| mpm_particle | 500,000 | gel             |
| mpm_grid     |         | field, grid 64  |

## activities

-  seed  
build the cell mesh
-  structural  
drive the surface at a set rate
-  lateral  
contact with a live surface  
the material update
-  exchange  
particle to grid  
grid to particle
-  field  
the grid solve

# A vesicle growing arms, with the purse-string tension restored



## entities

|             |         |                    |
|-------------|---------|--------------------|
| cell        | 2,000   | of 65,004 declared |
| vertex      | 130,004 |                    |
| ↔ half_edge | 520,032 | vertex → vertex    |

## activities



seed

build the cell mesh  
the initial morphogen



aggregate  
cell geometry from the mesh



rewire  
who neighbours whom  
T1 neighbour exchange



lateral  
morphogen diffusion between neighbouring cells  
autocatalytic reaction  
chemistry driven by curvature  
growth where the morphogen is high  
vertex-model shape energy  
purse-string line tension



divide  
cell division